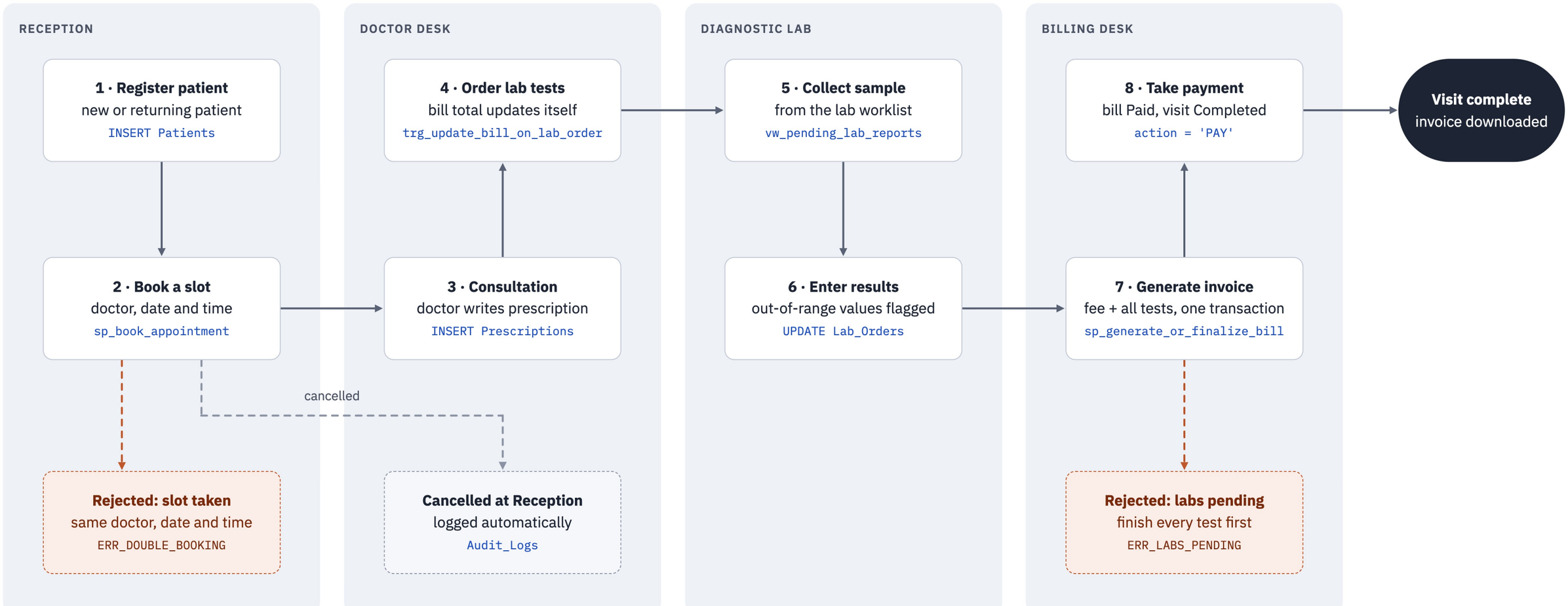
Patient journey: registration to payment

Each column is one tab of the Streamlit app. Steps run 1 to 8; dashed branches show what the database rejects or records.



Normal path — Rejected by a trigger or procedure — Alternative path — Blue `mono text` = the database object each step uses

Digital Health Clinic & Diagnostic Lab Portal

Connected to **clinic_portal** on localhost

Today, 17 Oct 2026

Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception, then Doctor Desk, Diagnostic Lab and Billing. All SQL uses %s placeholders.

Register new patient

First name * Ravi

Last name * Kumar

Date of birth * 12/03/1985

Gender * M

Phone * (10 digits) 9840020001

Email ravi.kumar@mail.com

Blood group B+

Address Anna Nagar, Chennai

Register patient — `INSERT Patients · UNIQUE phone, email`

Patient registered. Patient ID = 16

Book appointment

Search patient (name or phone) Ravi

Patient 1 - Ravi Kumar (9840020001)

Department Cardiology

Doctor Dr. Rahul Verma (fee 900.0

Date 18/10/2026

Time slot (free slots only) 10:30

☐ Show booked slots too (to demonstrate the double-booking error)

Reason for visit Chest discomfort

Book appointment — `CALL sp_book_appointment`

Doctor already has an appointment at this time (error state: trigger rejects a taken slot)

Upcoming appointments

ID	Date	Time	Patient	Doctor	Status
16	17 Oct 2026	10:00	Ravi Kumar	Dr. Rahul Verma	Scheduled
11	18 Oct 2026	10:00	Nisha Joseph	Dr. Arvind Menon	Scheduled
12	18 Oct 2026	11:00	Vijay Anand	Dr. Rahul Verma	Scheduled
13	19 Oct 2026	09:00	Pooja Krishnan	Dr. Anita Desai	Scheduled

Cancel an appointment

#11 - Nisha Joseph with Dr. Arvind Menon on 18 Oct 10:00

Cancel selected appointment — `trg_audit_appointment_cancellation`

Database objects on this screen: Patients, Departments, Doctors, Appointments · procedure sp_book_appointment · triggers trg_check_doctor_availability, trg_audit_appointment_cancellation

Digital Health Clinic & Diagnostic Lab Portal

Connected to **clinic_portal** on localhost

Today, 17 Oct 2026

Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception, then Doctor Desk, Diagnostic Lab and Billing. All SQL uses %s placeholders.

Doctor: **Dr. Rahul Verma (Interventional Cardiology)**

☐ Only this date

Date: 17/10/2026

ID	Date	Time	Patient	Status
16	17 Oct 2026	10:00	Ravi Kumar	Scheduled
12	18 Oct 2026	11:00	Vijay Anand	Scheduled

Open consultation: **#16 - Ravi Kumar at 10:00 (17 Oct 2026)**

Patient details

Name: Ravi Kumar **DOB:** 12 Mar 1985 · **Gender:** M **Blood group:** B+ · **Phone:** 9840020001

Reason for visit: Cardiac screening

› Previous prescriptions (2)

Prescription

› Medication not in the list? Add it

Medicine: **Atorvastatin 10mg**

Dosage: 1 tablet | Frequency: once daily | Duration (days): 30

Notes: At night

Add to prescription `INSERT Prescriptions`

Medicine	Dosage	Frequency	Days
Atorvastatin	1 tablet	once daily	30
Amlodipine	1 tablet	once daily	30

Order lab tests

Tests (choose several)

- Fasting Blood Glucose (Biochemistry, 80.00)
- Total Cholesterol (Lipid Profile, 300.00)
- HDL Cholesterol (Lipid Profile, 350.00)

Order selected tests `INSERT Lab_Orders · fires bill trigger`

Test	Status	Result	Unit	Cost
Fasting Blood Glucose	Ordered	-	mg/dL	80.0
Total Cholesterol	Sample Collected	-	mg/dL	300.
HDL Cholesterol	Ordered	-	mg/dL	350.

Cannot add lab tests to an appointment whose bill is paid (error state)

Database objects on this screen: Doctors, Appointments, Patients, Medications, Prescriptions, Diagnostic_Tests, Lab_Orders · triggers trg_update_bill_on_lab_order, trg_block_lab_order_after_payment

Digital Health Clinic & Diagnostic Lab Portal

Connected to **clinic_portal** on localhost

Today, 17 Oct 2026

Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception, then Doctor Desk, Diagnostic Lab and Billing. All SQL uses %s placeholders.

Pending tests	Waiting for sample	Waiting for result
4	3	1

Pending worklist

`SELECT * FROM vw_pending_lab_reports`

Order	Patient	Test	Status	Ordered by	Waiting
12	Ravi Kumar	Fasting Blood Glucose	Ordered	Dr. Rahul Verma	3 h
13	Ravi Kumar	Total Cholesterol	Sample Collected	Dr. Rahul Verma	3 h
14	Ravi Kumar	HDL Cholesterol	Ordered	Dr. Rahul Verma	3 h
15	Vijay Anand	ECG	Ordered	Dr. Rahul Verma	1 h

1. Collect sample

Order

Order #12 | Ravi Kumar | Fasting Blood Glucose

Mark sample collected

`status: Ordered to Sample Collected`

Sample collected for order #12 (Fasting Blood Glucose).

2. Enter result

Order

Order #13 | Ravi Kumar | Total Cholesterol

Reference range: 0.00 to 200.00 mg/dL

Result value (mg/dL): 225.00

Notes: Borderline high

Complete order

`status: Completed · bill recalculated`

Order #13 (Total Cholesterol) completed. Result is HIGH.

Abnormal results (outside reference range)

Patient	Test	Result	Range	Unit	Flag
Suresh Babu	Total Cholesterol	225.00	0 to 200	mg/dL	HIGH
Lakshmi Narayanan	HbA1c	6.10	4.0 to 5.6	%	HIGH
Aarav Patel	Hemoglobin	11.20	12 to 17	g/dL	LOW

Database objects on this screen: view vw_pending_lab_reports · Lab_Orders, Diagnostic_Tests · trigger trg_update_bill_on_lab_order (on UPDATE) · query Q8 abnormal results

Digital Health Clinic & Diagnostic Lab Portal

Connected to **clinic_portal** on localhost

Today, 17 Oct 2026

Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception, then Doctor Desk, Diagnostic Lab and Billing. All SQL uses %s placeholders.

Reception | Doctor Desk | Diagnostic Lab | **Billing & Analytics**

Billing Desk | Analytics

Visit to bill

#10 Harish Gupta with Dr. Meera Iyer (14 Oct 10:00) [Generated]

☐ Only visits ready to bill

Harish Gupta

Cardiology · Dr. Meera Iyer · visit #10 on 14 Oct 2026 at 10:00

Item	Status	Amount (₹)
Consultation - Dr. Meera Iyer	-	850.00
Lab - Triglycerides	Completed	350.00

Expected total: ₹1,200.00

Shown instead when a test is unfinished: 1 lab test is not completed yet, so the procedure refuses to invoice this visit.

Invoice

Regenerate invoice

`sp_generate_or_finalize_bill(10, 'GENERATE')`

Amount due

₹1,200.00

Payment mode

UPI

Mark as paid

`sp_generate_or_finalize_bill(10, 'PAY', 'UPI')`

Download invoice (.txt)

Recently paid

Bill	Paid at	Patient	Doctor	Consultation ₹	Labs ₹	Total ₹	Mode
8	16 Oct, 15:10	Kavitha Raj	Dr. Sunita Rao	500.00	400.00	900.00	Cash
7	16 Oct, 16:10	Sneha Reddy	Dr. Farhan Sheikh	550.00	0.00	550.00	UPI
6	16 Oct, 17:10	Mohammed Iqbal	Dr. Vikram Singh	700.00	900.00	1,600.00	Cash
5	16 Oct, 18:10	Aarav Patel	Dr. Karthik Subramanian	500.00	150.00	650.00	Card

Database objects on this screen: procedure sp_generate_or_finalize_bill (one transaction: sums fee + every test, writes Bills, marks the visit Completed) · function fn_lab_total · Bills, Appointments, Lab_Orders

Connected to **clinic_portal** on localhost

Today, 17 Oct 2026

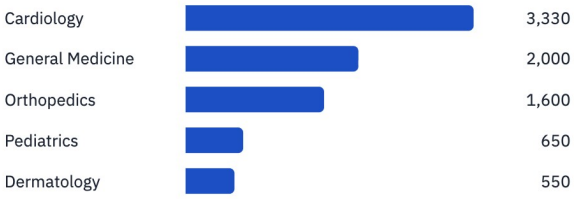
Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception, then Doctor Desk, Diagnostic Lab and Billing. All SQL uses %s placeholders.

Period: Last 7 days | **Last 30 days** | Last 90 days | Last 365 days

Revenue collected	Paid bills	Average bill	Visits	Outstanding (unpaid)
₹8,130.00	8	₹1,016.25	10	₹1,200.00

Revenue by department



Show SQL (JOIN Bills, Appointments, Doctors, Departments · GROUP BY)

Revenue per day



Hover a point for the day and amount.

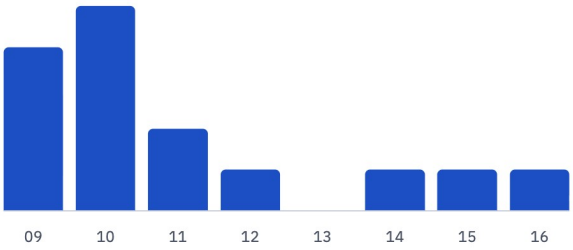
Show SQL (GROUP BY DATE(paid_at))

Most ordered lab tests



Show SQL (GROUP BY test · HAVING COUNT(*) >= 2)

Busy hours



Peak hour: **10:00** with 5 appointments.

Show SQL (GROUP BY HOUR(appt_time))

Doctor workload and revenue

Doctor	Department	Visits	Revenue ₹	₹ per visit
Dr. Rahul Verma	Cardiology	2	2,130	1,065
Dr. Arvind Menon	General Medicine	2	1,100	550
Dr. Vikram Singh	Orthopedics	1	1,600	1,600
Dr. Sunita Rao	General Medicine	1	900	900

Show SQL (LEFT JOIN, CASE, NULLIF)

Payment modes



Show SQL

› Audit trail (written by trg_audit_appointment_cancellation)

Database objects on this screen: Bills, Appointments, Doctors, Departments, Lab_Orders, Diagnostic_Tests, Audit_Logs · aggregate queries with JOIN, GROUP BY, HAVING, CASE · every chart is one SQL query shown under it